AgriIntelligence" – End-to-End Crop, Weather & Market Analytics Platform

DATA SCIENCE (24AD3107) Project Review-1

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Under the Guidance of

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Outline of Project Work

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Problem Statement

AgriIntelligence integrates crop yield, weather conditions, and agricultural market prices into a single analytics and decision-support platform.

- Agricultural datasets are often fragmented across different sources and formats.
- Manual comparison of yield, weather, and market information is time-consuming.
- The proposed platform combines data preprocessing, analytics, machine learning, weather-risk assessment, and crop recommendation.

Objectives

- Integrate crop yield, weather, and market price datasets.
- Clean, preprocess, and combine agricultural data.
- Perform exploratory analysis of yield, weather, and price trends.
- Develop a Random Forest model for crop-yield prediction.
- Develop a Random Forest model for market-price prediction.
- Assess weather conditions and classify agricultural risk.
- Generate crop recommendations using yield, price, and weather factors.
- Provide an interactive Streamlit dashboard.

AgriIntelligence

End-to-End Crop, Weather & Market Analytics Platform

From Data to Smarter Farming Decisions

1. DATA SOURCES

Crop Yield Data
(State, Crop, Year, Production, Area, etc.)

Weather Data
(Rainfall, Temperature, Humidity, etc.)

Market Price Data
(Commodity Prices, Market, Year, etc.)

2. DATA UNDERSTANDING

Dataset inspection and initial analysis

3. DATA PREPROCESSING

- Missing values
- Duplicate removal
- Data type conversion
- Data validation

4. DATA INTEGRATION

Merge crop, weather and market data

5. EXPLORATORY DATA ANALYSIS

Trends, patterns and relationships

6. FEATURE ENGINEERING

Create ML-ready features

7. YIELD PREDICTION

Random Forest

Predict crop yield based on integrated data

8. MARKET-PRICE PREDICTION

Random Forest

Predict market price using historical and temporal information

9. WEATHER-RISK ANALYSIS

Assess weather risk based on rainfall, temperature and humidity

Low | Moderate | High

10. CROP RECOMMENDATION

Recommend best crops using yield potential, market price and weather suitability

11. STREAMLIT DASHBOARD

Interactive web application for visualization and decision support

12. OUTPUT: ANALYTICS + PREDICTIONS + RISK + RECOMMENDATIONS

Crop & Weather Analytics | Yield Predictions | Market Price Predictions | Weather Risk Assessment | Crop Recommendations

Literature Survey

DEEMED TO BE UNIVERSITY

Authors	Title of the Paper	Methodology	Inferences	Limitation/Future work	Our Implementation
van Klompenburg et al. (2020)	Crop Yield Prediction Using Machine Learning: A Systematic Literature Review	• Review of ML approaches • Analysis of datasets, input variables and evaluation methods used for crop-yield prediction	ML can combine environmental and agricultural variables for yield prediction.	Performance depends on data quality, geographical coverage, selected features and model choice.	We integrate crop, weather and location information, perform preprocessing and feature engineering, and build a complete yield-prediction pipeline.
Khaki & Wang (2019)	Crop Yield Prediction Using Deep Neural Networks	• Deep neural networks using large maize datasets with historical crop and environmental information	Shows the potential of data-driven models for crop-yield prediction.	Requires large and diverse datasets and substantial modelling/tuning for better generalization.	We use Random Forest as an initial tabular-data model and evaluate it using MAE, RMSE and R^2.
Crane-Droesch (2018)	Machine Learning Methods for Crop Yield Prediction and Climate Change Impact Assessment in Agriculture	• ML methods applied to crop-yield prediction and climate-related variables • Use of environmental and agricultural data	ML can model nonlinear relationships between crop yield and climate.	Future work can incorporate richer weather, soil and remote-sensing data to improve prediction accuracy.	We incorporate rainfall, temperature and humidity data and add a separate weather-risk analysis module.
Theofilou et al. (2025)	Predicting Prices of Staple Crops Using Machine Learning: A Systematic Review of Studies on Wheat, Corn, and Rice	• Review of ML, ensemble learning, deep learning and time-series approaches for agricultural commodity price prediction • Analysis of datasets and evaluation metrics	ML models can capture price trends and can incorporate historical prices along with other factors such as weather, production and economic information.	Challenges include data availability, region-specific datasets, model interpretability and computational complexity.	We implemented a market-price prediction module using historical price and temporal information, and integrated it into the dashboard with price analytics and prediction visualizations.
Integrated agricultural decision-support direction	From Prediction to Decision Support in Agriculture (General direction from multiple studies)	• Existing research demonstrates prediction of individual agricultural variables (yield, weather or price) • Different components are often handled separately	Combining multiple agricultural factors can provide better insights for decision-making.	Prediction alone does not provide a unified user-facing decision-support workflow. Integration of analytics, recommendation and visualization is limited.	We combine yield prediction + price prediction + weather risk + crop recommendation in one integrated workflow and provide it through an interactive Streamlit dashboard.

Problem Gap Identification

- Existing agricultural studies often focus on one factor such as crop yield or weather.
- Crop, weather, and market-price information is not always available through one integrated decision-support interface.
- Analytics, prediction, risk assessment, and recommendation are frequently treated as separate components.
- AgriIntelligence addresses this gap by integrating these components into a single workflow and dashboard.

Innovation, Creativity, and Novelty

- Unified integration of crop yield, weather, and market-price information.
- End-to-end pipeline from data cleaning and integration to ML prediction and visualization.
- Dual prediction capability for crop yield and market price.
- Weather-risk assessment using rainfall, temperature, and humidity.
- Location- and year-aware crop recommendation using yield, price, and weather suitability.
- Interactive Streamlit dashboard for analytics, predictions, and recommendations.

Future Improvements

🌱 **Real-World Agricultural Data**
Integrate larger and continuously updated datasets from government and agricultural sources.

 Soil & Satellite Data Integration
Incorporate soil nutrients, soil moisture, vegetation indices and satellite/remote-sensing data.

 Advanced ML Models
Compare Random Forest with XGBoost, Gradient Boosting, LSTM and other models to improve prediction performance.

 Weather Forecast Integration
Use real-time and forecast weather data for early risk detection and proactive recommendations.

💰 **Profit & Revenue Prediction**
Extend price and yield predictions into expected revenue/profit estimation for different crop choices.

 Geospatial Agricultural Analytics
Add district-level maps and heatmaps for crop suitability, yield, price and weather risk.

📱 **Mobile & Alert System**
Develop a mobile-friendly version with notifications for price changes, weather risks and crop recommendations.

 Explainable AI Recommendations

Feasibility and Project Plan

The project is feasible using Python, Pandas, Scikit-learn, Matplotlib, Joblib and Streamlit. The prototype uses structured agricultural datasets and can be extended with larger real-world datasets.

Phase 1 — Data collection & understanding

Phase 2 — Data cleaning & preprocessing

Phase 3 — Data integration & exploratory analysis

Phase 4 — Feature engineering & ML model development

Phase 5 — Weather risk & crop recommendation

Phase 6 — Streamlit dashboard integration

Phase 7 — Testing, documentation & deployment